

Math 241, F18
Quiz 10

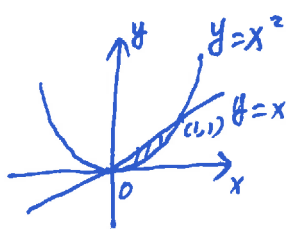
Name: Answer

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz.

Problem 1. (5 points) Evaluate $\int_0^1 \int_0^{y^2} 3y^3 e^{xy} dx dy$

$$\begin{aligned} & \int_0^1 \int_0^{y^2} 3y^3 e^{xy} dx dy \\ &= \int_0^1 [3y^3 e^{xy}]_{x=0}^{x=y^2} dy \\ &= \int_0^1 3y^3 e^{y^3} - 3y^3 dy \\ &= [e^{y^3} - y^3]_{y=0}^{y=1} \\ &= [e-1] - [1-0] = e-2 \end{aligned}$$

Problem 2. (5 points) Evaluate $\iint_R x+2y dA$, where R is the region enclosed by $y=x$ and $y=x^2$.



$$\begin{cases} y=x^2 \\ y=x \end{cases} \Rightarrow x^2=x, \quad x(x-1)=0 \\ x=0, x=1 \\ y=0 \quad y=1 \end{aligned}$$

$$\Rightarrow R = \{(x,y) \mid 0 \leq x \leq 1, x^2 \leq y \leq x\}$$

$$\begin{aligned} \iint_R x+2y dA &= \int_0^1 \int_{x^2}^x x+2y dy dx \\ &= \int_0^1 [xy + y^2]_{y=x^2}^{y=x} dx \\ &= \int_0^1 [x^2 + x^2] - [x^3 + x^4] dx \\ &= \int_0^1 2x^2 - x^3 - x^4 dx \\ &= \left[\frac{2}{3}x^3 - \frac{x^4}{4} - \frac{x^5}{5} \right]_{x=0}^{x=1} \\ &= \frac{2}{3} - \frac{1}{4} - \frac{1}{5} = \frac{13}{60} \end{aligned}$$